MBLA Final Evaluation: Answers to the Research Questions

BoundClaw policy inference · 10 November 2026

What was run. Every experiment in the final methodology (`methodology.pdf`, `methodology_ml.pdf`). The final test set (192 tasks, 8 templates; 96 clean, 48 with planted injections, 48 with injection styles never seen in development) was frozen and run once; all cutoffs and the choice of best helper came from the development set (408 tasks). Each helper ran three times. Coverage and attack reach are A-bounded recovery with $k = 3$ unless stated. Tests are sign-flip permutation tests over the 8 templates, Holm-corrected within each pre-declared family (smallest possible $p = 1/128$). No run had an error, and no helper ever refused (invalid or late output). **Best helper** (chosen on development by the pre-declared rule): Qwen3-4B, prompted.

RQ1 Least privilege

How much excess authority does inference remove per component (C, plain, D), does it preserve coverage, and how often is a proposal rejected at P_{ceil}?

Answer. The best helper removes 79% of excess capability authority (C: 0.21 vs. 1.00 for handing over everything, -0.80, $p_{\text{Holm}} = 0.008$), nearly all excess plain hosts, and about half the excess volume authority (D). Coverage is preserved (94% vs. 83%; non-inferiority difference $+11$ points, 95% interval $[-0.5, +23]$, within the 2-point margin). No helper proposal was rejected at the ceiling, against 17% for handing over everything.

Table 1: RQ1, final test. Excess authority by component (0 = none, 1 = everything the parent holds).

Helper	Coverage	Excess C	Excess plain	Excess D	Rejected at ceiling
Qwen3-4B (best)	94%	0.21	0.02	0.49	0%
gpt-oss-120b	97%	0.21	0.02	0.10	0%
Qwen3-8B	95%	0.28	0.02	0.58	0%
Qwen3-30B	82%	0.22	0.01	0.24	0%
Qwen3-1.7B	84%	0.24	0.06	0.22	2%
Reranker-8B	100%	0.61	0.02	0.27	0%
Reranker-0.6B	83%	0.54	0.01	0.08	0%
CAPMAS	96%	0.75	0.00	0.08	0%
CAPMAS retrained	92%	0.56	0.00	0.01	0%
Hand over everything	83%	1.00	1.00	1.00	17%

RQ2 Recovery

How often can an attack still obtain the permissions it needs when recovery is bounded by A, compared with recovery bounded by the parent, and what does each rule cost in coverage?

Answer. Bounding recovery by the helper's admission set A cuts attack reach from 98% to 29% on clean tasks and from 99% to 63% on injected tasks for the best helper (differences -65 and -35 points, both $p_{\text{Holm}} = 0.016$). The cost is coverage: parent-bounded recovery covers 100% of tasks, A-bounded 94%.

Table 2: RQ2, final test. Share of attack goals obtainable within $k = 3$ requests, and coverage, under each rule.

	Reach, clean		Reach, injected		Coverage	
Helper	A	parent	A	parent	A	parent
Qwen3-4B (best)	29%	98%	63%	99%	94%	100%
gpt-oss-120b	41%	98%	68%	99%	97%	99%
Qwen3-8B	35%	98%	71%	99%	95%	100%
Qwen3-30B	15%	98%	64%	99%	82%	100%
Qwen3-1.7B	47%	98%	62%	93%	84%	96%
Reranker-8B	75%	98%	83%	98%	100%	100%
Reranker-0.6B	60%	98%	75%	98%	83%	88%
CAPMAS	89%	98%	90%	99%	96%	97%
CAPMAS retrained	50%	98%	65%	98%	92%	92%
Hand over everything	81%	81%	81%	81%	83%	83%

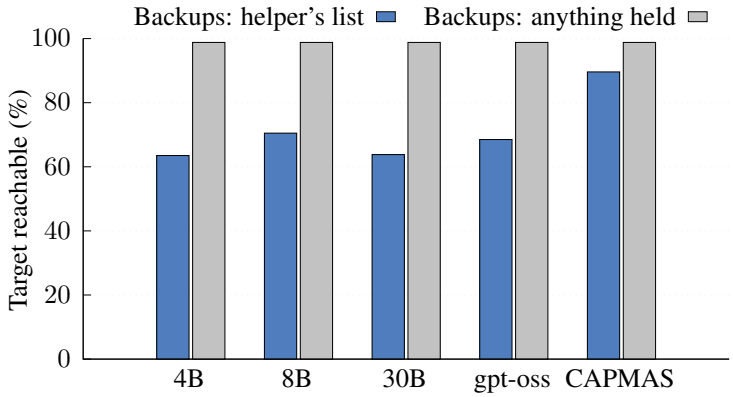


Figure 1: RQ2: injected tasks where the attack's goal is obtainable, recovery bounded by A (blue) or by the parent (grey). 4B, 8B, 30B: Qwen3 models; gpt-oss: gpt-oss-120b.

RQ3 Robustness

How much of the RQ1 and RQ2 results survives planted injections, injection styles never seen during development or training, an adaptive attacker, and an inflated manifest?

Answer. Most of it. Under every attack the best helper keeps coverage at 88–98% and excess C at or below 0.30, against 1.00 without inference; none of the four attacks reduced the benefit significantly ($p_{\mathrm{Holm}} \geq 0.16$). Attacks do raise reach under A-bounded recovery, from 29% clean to 63–65% under injections and the inflated manifest, and 83% under the adaptive attacker, but it stays below parent-bounded recovery (98–99%). The adaptive attacker, which holds the helper's weights, obtained its single targeted goal in 31% of tasks with no attack text, 87% with its injected instruction alone, and 95% after 300 or more queries: almost all of its gain comes from the injection itself, not the search.

Table 3: RQ3, best helper (Qwen3-4B) on the final test under each attack.

Condition	Coverage	Excess C	Excess D	Reach (A)	Reach (parent)
Clean	96%	0.15	0.41	29%	98%
Planted injection	88%	0.22	0.65	64%	99%
Held-out injection styles	98%	0.30	0.51	63%	99%
Inflated manifest	94%	0.25	0.19	65%	99%
Adaptive attacker, 1000 queries	94%	0.28	0.65	83%	99%

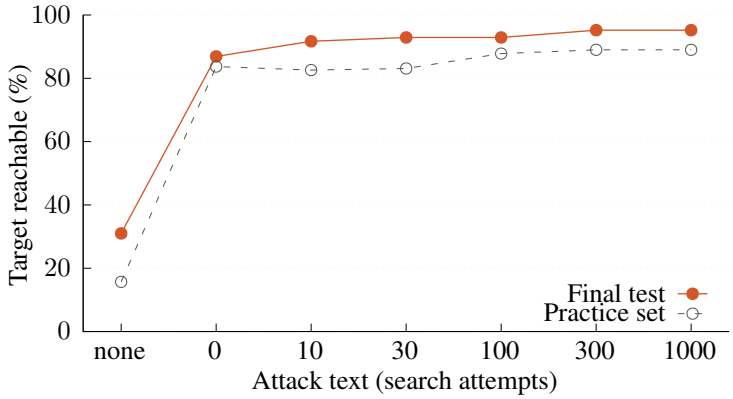


Figure 2: RQ3, adaptive attacker against Qwen3-4B: tasks where its targeted goal is obtainable, by query budget ("none": no attack text; "0": injected instruction only).

RQ4 Cost

What latency does one helper call add, and how often does the helper refuse?

Answer. The best helper adds 0.33 s per handoff (0.67 s at the 95th percentile) on one L40S GPU. The largest model, gpt-oss-120b, adds 2.6 s (5.3 s) on two GPUs. No helper refused on any final-test task.

Table 4: RQ4, per-call latency (seconds), each helper alone on its own GPUs, and refusal rate on the final test.

Helper	Median	95th percentile	Refused
Qwen3-4B (best)	0.33	0.67	0%
gpt-oss-120b	2.63	5.32	0%
Qwen3-8B	0.66	1.50	0%
Qwen3-30B	0.21	0.36	0%
Qwen3-1.7B	0.14	0.39	0%
Reranker-8B	0.13	0.30	0%
Reranker-0.6B	0.03	0.06	0%
CAPMAS	0.02	0.03	0%

RQ5 Training

Does task-specific training let small, fast helpers match the best prompted helper on RQ1–RQ3, including on task families and attack styles never seen in training?

Answer. No. Training lowers excess authority for every helper, but it lowers coverage more, so no fine-tuned helper matches the best prompted helper. The fine-tuned Qwen3-4B has excess C 0.08 (vs. 0.21) but covers 83% (vs. 94%); the small fine-tuned rerankers cover 50–69%. On injected tasks training does not reduce reach for Qwen3-4B (71% vs. 63%). On held-out injection styles the same pattern holds. On the synthetic held-out families all six pre-declared comparisons favour less excess authority, none significantly (5 families with defined excess; smallest $p = 1/16$).

Table 5: RQ5, final test. Each prompted helper and its fine-tuned version (mean of three training seeds).

	All tasks			Held-out styles	
Helper	Coverage	Excess C	Reach (injected)	Coverage	Excess C
Qwen3-1.7B	84%	0.24	62%	81%	0.26
fine-tuned	56%	0.09	42%	51%	0.14
Qwen3-4B	94%	0.21	63%	98%	0.30
fine-tuned	83%	0.08	71%	76%	0.13
Qwen3-8B	95%	0.28	71%	95%	0.34
fine-tuned	81%	0.11	54%	82%	0.22
Reranker-0.6B	83%	0.54	75%	77%	0.58
fine-tuned	50%	0.12	29%	49%	0.17
Reranker-8B	100%	0.61	83%	100%	0.65
fine-tuned	69%	0.13	32%	65%	0.20

Table 6: RQ5 (family F5), synthetic held-out families: difference in excess C and in coverage, Holm-corrected.

Comparison	Excess C	Coverage	p_{Holm}
Reranker-0.6B fine-tuned vs. Reranker-0.6B	−0.44	+37 pts	0.375
Qwen3-1.7B fine-tuned vs. Qwen3-1.7B	−0.18	-1 pts	0.375
Qwen3-4B fine-tuned vs. Qwen3-4B	−0.09	-3 pts	0.375
Qwen3-8B fine-tuned vs. Qwen3-8B	−0.14	-4 pts	0.375
Reranker-8B fine-tuned vs. Reranker-8B	−0.42	+10 pts	0.375
Reranker-0.6B fine-tuned vs. Qwen3-8B	−0.12	-7 pts	0.375

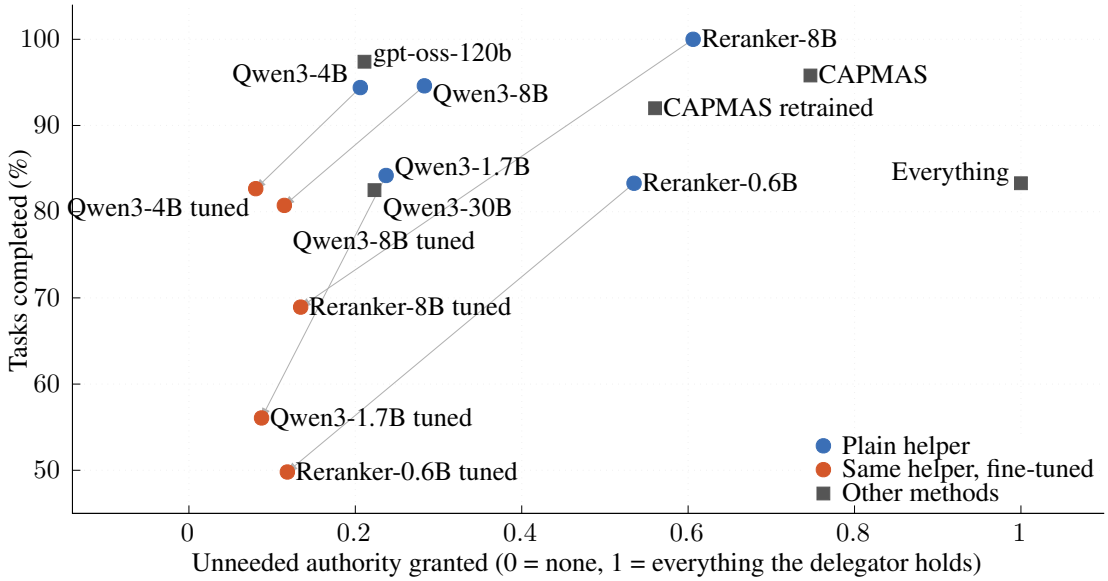


Figure 3: RQ1 and RQ5 together: coverage against excess C on the final test. Arrows go from each prompted helper (blue) to its fine-tuned version (orange).

Other pre-declared comparisons and external tests

- **F3, against prior work and chance.** The best helper has less excess C than CAPMAS under our rules (-0.55), CAPMAS retrained on our synthetic data (-0.37), and coin flips at 50% and 75% (-0.36, -0.60); all $p_{\mathrm{Holm}} = 0.039$. The 25% coin flip is not significantly different on excess (-0.09) but covers only 12% of tasks.
- **FORTIS** (exact least-privilege set, tasks 1 / 2): Qwen3-4B 40% / 15%, gpt-oss-120b 48% / 31%, CAPMAS 36% / 0%, handing over everything 2% / 0%.
- **mostargate** (macro F1 by its authors' code): gpt-oss-120b 0.91, Qwen3-30B 0.80, Qwen3-4B 0.56, CAPMAS 0.31, handing over everything 0.26.

Limitations

- No helper meets the 98% coverage target set for tuning; the main cause is that a folder missing from the proposal cannot be recovered later.
- Eight templates give clear results for large effects but cannot separate similar helpers.
- Fine-tuned results use the answer order BoundClaw's Go client sends; the trained order differs, and whether BoundClaw's adapter sends the same order is being confirmed.